Topic 1 Explained Simply: Superimposition

A walk through the colleague's summary · Doppel 1994 · Nielsen 1989 · Nguyen 2018

🧩 The one question behind all three papers

A child comes in at 11 and again at 15, and the teeth have moved. **Did the braces move them, or did the face grow and carry them?** You can't tell from one X-ray, so you lay the two X-rays on top of each other and **line them up on a bone that did not change**. Whatever still looks different is real change. That lining-up is **superimposition**.

The catch: a growing face is not frozen. Some surfaces add bone, some lose bone, and whole jaws drift. Line up on a bone that moved and you hide real change or invent fake change. So the whole topic is one question: **which bones can you trust to stay still?**

How they found out: Björk put tiny **metal pins (implants)** into children's jaws. Pins don't grow, so lining up the pins shows the truth. Then you check which natural bones stayed lined up with the pins.

🦴 Paper 1 · Doppel et al. 1994: which upper-jaw spots stay still?

What they did: 50 children with three pins in the upper jaw, X-rays about **4 years apart** (11.9 → 16.0 years). They lined up the pins, then measured **how far each of 17 natural spots drifted away**. Small drift means trustworthy.

How to read the colleague's table: "mean displacement" = **how far the spot drifted**, in mm. Think of it as a trust score: **about 1 mm or less = good**; everything above that was rejected, even the palate borders at 1.05–1.14 mm.

- **Zygomatic process (cheekbone ridge), front and back: about 1 mm.** The winners. The back surface can **always** be traced and always moves the same way (backward, as bone is added). The front can be traced in 80 % of films and moves forward in some children (27.5 %), backward in others (35 %) and not at all in the rest (37.5 %).
- **Internal palate: 0.85 mm, the smallest number, but rejected.** It *looked* wrong when lined up by eye, so a good number alone wasn't enough.
- **ANS, PNS, A point, PTM: 1.3–2.4 mm.** Don't register on them: the front of the palate sinks, the back grows backward, and A point moves with the teeth.

The rotation table: the **palatal plane** (ANS–PNS line) tilted anywhere from **−5° to +8°** inside the maxilla. A line that tilts differently in every child **can't be your reference line**.

The 1.5 : 1 rule: as the child grows, the **eye-socket floor rises 1.5 mm for every 1 mm the palate sinks**. So when you line up on the cheekbone, let the orbit floor rise a bit more than the nasal floor falls (Björk and Skieller found 3 : 2, the same ratio).

🔀 Paper 2 · Nielsen 1989: three ways to line up the upper jaw

What they did: 18 children with pins, X-rays 4 years apart, lined up three ways:

1. **Best fit:** line up the palate outlines at ANS. The old, easy way.
2. **Implants:** the truth.
3. **Structural (Björk–Skieller):** line up on the front of the cheekbone ridge and let the orbit floor rise as much as the nose floor sinks.

The result in one sentence: best fit **hides about 30 % of molar eruption and 50 % of incisor eruption**, because the palate itself sinks and drags the reference down with it. The structural method matched the pins vertically.

How to read the colleague's numbers table (how far the teeth moved down in 4 years):

	Molar cusp	Incisor edge
Implants (truth)	4.75 mm	3.39 mm
Best fit	3.40 mm	1.40 mm
Structural	4.99 mm	3.39 mm

Best fit saw the incisor come down **1.40 mm**; really it came down **3.39 mm**. That missing 2 mm is the sinking palate. The paper does not say exactly how it got 30 % and 50 %. From its Table II, best fit misses about 28 % of the molar cusp's movement and about 59 % of the incisor edge's (derived).

🧊 Paper 3 · Nguyen et al. 2018: the lower jaw in 3D

What they did: Björk's pin studies can't really be repeated today (radiation-dose and ethics-board rules), so they used **bone plates** already placed for Class III treatment as the "pins". 20 growing children had CBCT scans 1 year apart. A computer lined up the two scans on the plates and screws (voxel-based registration), then measured how far each of three candidate spots had moved.

- **Chin front: 0.37 mm. Inside of the chin bone (symphysis): 0.40 mm.** Stable, under half a millimetre.
- **Wisdom-tooth pocket (third-molar crypt): 1.94 mm.** It moves, so don't use it.
- **In a separate group of **25 growing children** (20 Class II, 5 Class III), **two different people** each lined up the scans on chin + symphysis. Their results differed by only **0.12 mm on average**, and the **ICC was 0.998**, near perfect.

📖 What is what

Term	Meaning
Superimposition / registration	Stacking two X-rays; the bone you line them up on is where you "register"
Implants (here)	Tiny metal marker pins in bone, not dental implants
Zygomatic process, key ridge	The curved cheekbone root seen above the upper first molar
Orbital floor / nasal floor	Bottom of the eye socket (gains bone, rises) / top of the palate (loses bone, sinks)

Term	Meaning
ANS / PNS / PTM / A point	Front tip of the upper jaw / back end of the palate / teardrop gap behind the maxilla / deepest point above the upper incisors
Symphysis	The chin bone where the two halves of the lower jaw join
CBCT, voxel	3D X-ray; a voxel is a 3D pixel (0.3 mm here)
ICC	Agreement between two people; 1.0 = perfect

🔎 Small points to know about the colleague's summary

- **"Posterior zygomatic process most stable":** the two cheekbone numbers are practically equal (0.97 vs 0.95 mm). Doppel does not rank one surface above the other: the abstract and conclusions recommend both. If you favour the back surface, the reason is that it can **always** be traced and always moves the same way, not its number (which is actually slightly larger).
- **"Infraorbital foramen visible in only 32 %":** that is what Doppel's text says, but the paper's own table lists it in 34 of 50 children (68 %). The paper contradicts itself.
- **Nielsen's implant molar value:** the colleague's numbers table gives 4.15 mm, but Nielsen's Table II prints **4.75 mm**. Use 4.75; with it, best fit misses about 28 % of the molar's movement.
- **"Björk and Skieller disputed this conclusion":** in the paper this sentence is about Moss and Greenberg's infraorbital-foramen method; in the summary it has lost that context.

🎯 If a consultant asks…

- **"Why not line up on the palate? It's right there."** Because the palate sinks as the child grows: Nielsen showed it hides 30–50 % of tooth eruption over 4 years.
- **"Where do you register the maxilla?"** On the front and back of the cheekbone ridge. Allow for the orbit floor rising about 1.5 times what the palate sinks (Doppel); Nielsen's version of Björk and Skieller's structural method set the two equal and slightly overstated eruption.
- **"And the mandible?"** On the chin and the inside of the symphysis; avoid the third-molar crypt (Nguyen: 0.37–0.40 mm vs 1.94 mm).
- **"What's weak about these studies?"** Small, selected samples (Doppel used 50 of about 700 implant records; Nielsen 18 of 35), treated or mixed samples (Nielsen mixed treated and untreated children and did not separate them), and in 3D only one year of growth.